

EarthScape User Guide

1. Introduction

Welcome to the EarthScape Climate Monitoring System. This platform utilizes **Big Data technologies** and **Machine Learning** to track global temperature trends, detect anomalies, and predict future climate conditions.

This guide provides step-by-step instructions for Climate Analysts and System Administrators to use the application effectively.

2. Getting Started

2.1 Accessing the Application

1. Open your web browser (Chrome, Firefox, or Edge).
2. Navigate to the local server URL: `http://127.0.0.1:5000/`.
3. You will land on the Home Page, which displays global climate statistics and the system status.

2.2 Creating an Account

1. Click the Sign Up button on the top right.
2. Username: Enter a unique username (e.g., `analyst_01`).
3. Password: Enter a secure password.
4. Role Selection:
 - Climate Analyst: Select this to access the Prediction Dashboard.
 - Administrator: Select this to manage Data Ingestion (requires authorization).
5. Click Create Account. You will be redirected to the Sign In page.

2.3 Logging In

1. Click Sign In on the navigation bar.
2. Enter your credentials.
3. Upon success, you will be redirected to the Dashboard (Analysts) or Ingestion Panel (Admins).

3. For Climate Analysts: Using the Dashboard

The Analyst Dashboard is the command center for viewing historical data and running predictive models.

3.1 HDFS Status Check

Look for the status indicator at the top right of the dashboard:

-  Online (Active): The HDFS cluster is connected and ready.
-  Offline: The system is running in simulation mode.

3.2 Running a Climate Prediction

1. Select Region: Use the dropdown menu to choose a major city (e.g., *London*, *Tokyo*, *New York*).
2. Forecast Period: Click the calendar icon to select a specific Month and Year (e.g., *July 2030*).
3. Click the Run Prediction button.

3.3 Interpreting Results

- Prediction Card: A card will appear showing the predicted average temperature (e.g., *24.5°C*) for that specific date.
- Trend Graph: Below the result, a line graph will generate automatically. This visualizes the temperature trend for the entire selected year, helping you spot seasonal patterns.
- Download: You can right-click the graph to save it for reports.

4. For Administrators: Data Ingestion

Note: The "Ingestion" tab is strictly visible only to users with the Administrator role.

4.1 Uploading New Datasets

1. Navigate to the Ingestion tab in the top navigation bar.
2. Locate the "Upload New Dataset" card.
3. Click Choose File and select a standard climate CSV file (e.g., `GlobalLandTemperatures.csv`).
4. Click Start Ingestion.

4.2 Monitoring Jobs

- Once uploaded, the system simulates a Hadoop Job submission.
- A "Success" badge will appear in the Recent Ingestion Jobs list on the right, confirming the data is now stored in the `/user/earthescape/climate_data/` HDFS directory.

5. Troubleshooting & Support

5.1 Common Issues

Issue	Possible Cause	Solution
"Prediction Failed"	Model files missing.	Contact IT to run <code>training/train_model.py</code> .
"Access Denied"	Wrong User Role.	Ensure you are logged in as an Admin to access Ingestion.
Graph not showing	Browser Cache.	Refresh the page (Ctrl + F5) to clear old assets.

5.2 Contact Support

If you encounter a persistent error:

1. Click Feedback in the navigation bar.
2. Fill out the support form with your Issue Type and Description.
3. Our engineering team will receive an automated email alert.